SWaP: A Water Process Testbed for ICS Security Research

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***Abstract*—This paper presents the Security Water Processing (SWaP) testbed, an Industrial Control System (ICS) for security research and training. SWaP will be used to (a) assess the security of industrial protocols, (b) design and test the defense technologies, (c) assess the effectiveness of side channel based defenses, and (d) share the datasets with the security community. SWaP consists of a 3-stage water process, each stage is autonomously controlled by Programmable Logic Controllers (PLCs). The local communications between sensors, actuators, and PLCs is realized through wired and wireless channels. All the details on the process layer, network layer and field layer are provided in this paper. The testbed is designed to be a resource to a wider research community and encourage collaborative research in the field of ICS security. We envision to make this resource available for practical work among the ICS security community. Towards that end, as a first step, we have collected and curated a dataset to be shared with the research community. Moreover, given the scarcity of attack/anomalous data; several attacks are designed and executed on the testbed, providing examples of attack data, which can further be used for model based and machine learning based security research.**

***Index Terms*—Cyber physical systems, Industrial control systems, Security, Testbed**

I. Introduction

Recent progress in technology is resulting in the digitization of the physical world and things around us. It is expected that communication and computing capabilities will soon be part of all physical objects [10]. The integration of computing with the physical world gives rise to complex systems referred to as Cyber Physical Systems (CPS).Some examples of CPS are manufacturing, transportation, power utilities, medical devices and Industrial Control Systems (ICS) [2], [11].

CPS is a broad term; in this work, we focus on Industrial Control Systems (ICS). In particular we consider an example of a water processing system. It is composed of water treatment, distribution and end-users as different entities. As one can imagine these water systems are composed of a multitude of devices and physical processes. Water treatment and distribution depend on the demand from the utilities and the users. To meet the requirements of the water demand the critical infrastructure is utilized to ensure a continuous supply of water. Each of the processes in the critical infrastructure is a complex engineering system and needs a sophisticated control to achieve its desired objectives. For example, at the treatment stage, we have filtration process and chemical dosing, all these devices are autonomously controlled by the Programmable Logic Controllers (PLC). This means that we have a lot of sensors monitoring the physical process, actuators and the physical infrastructure that communicate the current physical states with each other and with the PLC. Such communication among devices, on one hand, provides flexibility in controlling the complex CPS but increases the attack surface.

Successful attacks on CI have led to a surge in the research initiatives to defend against such attacks, however, there is a need to have facilities to test the possible defensive technologies. This is where testbeds are quite helpful. This report describes an ICS testbed, the Security Water Process (SWaP) testbed. SWaP is designed and built to enable experimental research in the design of secure ICS. This testbed is at core of the activities within the security groups at two UK Universities with a plan to extend the consortium. In the rest of the document we summarize the design and development of the testbed.

II. SWaP Testbed Introduction

Overview of SWaP testbed is shown in Figure 1. SWaP testbed is a three stage water process. Stage1 represents a secondary storage for the treated water, where as stage2 represents a primary storage. Stage3 represents consumer units. T1, T2, and T3 are the tanks in the respective stages of the testbed. Demand for users can be set dynamically and to fulfill the demand at first water is pumped from Stage2, the primary storage and if there is not enough water available to meet the demand, we shall pump water from Stage1, i.e., the secondary storage. Each stage is equipped with a set of sensors to measure the physical process, namely, level sensor L1 and flow sensor F1 in stage 1. V1 and V2 represent motorized valves in stage1 and P1 shows the pump. Stage3 consumer demand can be fulfilled either from stage2 and/or from stage1. Each stage is equipped with a 30L tank to hold the water. Since stage3 emulates the consumers and we do not really

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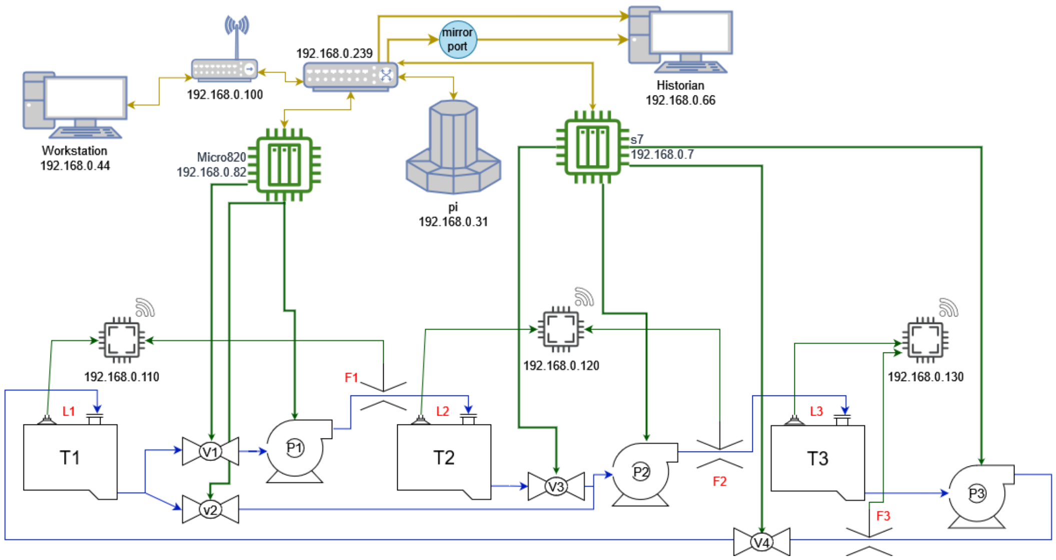

Fig. 1. SWaP testbed layout. It shows the key components at all levels. At field level, actuators and sensors are providing data to respective digital interface, which then communicate to a central switch. PLcs communicate through this central switch and historian keep s record of all the process data.

have the consumption of water, we recycle the same water back to stage1.

Figure 1 shows the network architecture of the testbed. At the lowest level we have signals transmitted between the PLCs and the field devices. Actuators are controlled directly by the PLCs based on the design parameters and sensor measurements. However, the sensor readings are collected at a central point before being transmitted to the PLCs. The PLCs can utilise the same network to communicate with each other by exchanging messages and they can be controlled and programmed through the Workstation and the data can be saved to a Historian process through the level2 network.

A. Testbed Design and Construction

The testbed, Figure 2, consists of three stages with each stage centred around a 30L water tank with an array of sensors, valves and a pump. Each tank has an ultrasonic level sensor mounted in the top of it which measures the current water level in the tank. Each tank has an inflow and outflow pipe with a pump mounted on the outflow and an inline flow sensor. Stage 2 and 3 have one valve each controlling the outflow to the next stage tank and stage 1 has two valves so it can supply to both stage 2 and 3 independently. The separation into stages also allows a separation of responsibilities with stage1 controlled by PLC1 and stages2 and 3 controlled by PLC2. A number of safety concerns had to be taken into account when designing the testbed as the combination of electricity and water in a lab environment has obvious implications. To that end we limit the use of AC devices to a single PLC and a standalone power supply which are kept isolated in a metal enclave and only use DC pumps and valves. We deployed two PLCs from different vendors to create a diverse environment, ultimately generating interesting interaction among devices. The Siemens S7-1200 PLC (PLC2) runs on 230v AC and provides an output of 24v DC, the standalone power supply provides 12v DC. These two DC sources are used to power the pumps, valves and the Allen-Bradley micro820 PLC (PLC1) . The Arduino boards used as the sensor interfaces are powered by USB.

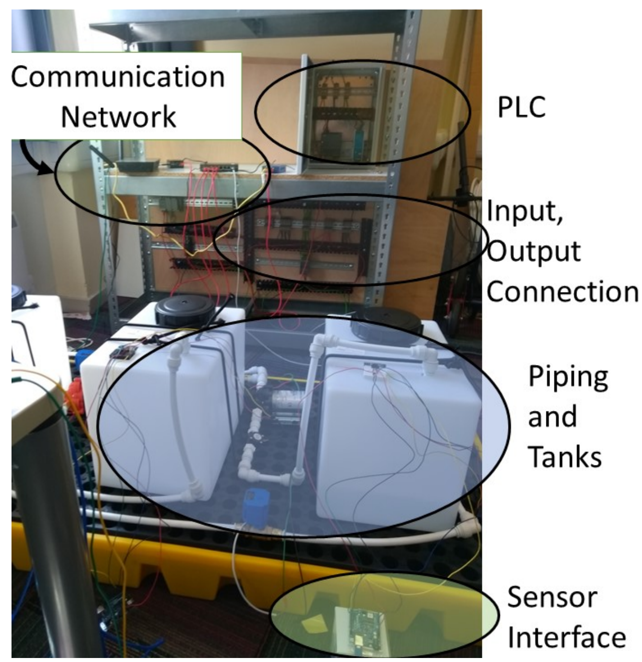

Fig. 2. A pictorial view of the SWaP testbed; showing Programmable Logic Controllers (PLCs), water tanks, sensors, wiring connections, actuators.

A major motivator for the testbed project was that it be modular and easy to build and fix which lead to us using PVC pipes and quick connect fittings for almost all of the plumbing. The pipes are easy to cut to the correct size and can be disconnected and repositioned while still being water tight. Currently all the actuators in the system are identical models but the modularity of the system makes it simple to swap them out whether for process variety or simply repairs.

B. Network Infrastructure

Network communications are an integral part of the testbed as the different system components are orchestrated over the network by the workstation and all the sensor data is transfer over the network. The testbed sits on its own network segment with the workstation PC also having internet access but does not bridge this access into the testbed network. The workstation, PLC1, PLC2, control unit and the historian are connected together through a switch. Each set of sensors is connected to an Arduino which are connected wirelessly to the network through a wifi access point.

Each Arduino regularly sends sensor readings to the control unit using a simple UDP payload containing an ID and the calculated sensor values. The control unit then directly writes these values to memory locations on the PLCs. The Siemens s7-1200 uses the s7-comm protocol and the Allen-Bradley micro820 uses EtherNet/IP . Other methods and industrial protocols could be used but these were chosen for ease of integration and availability of programming libraries. Figure 1 shows the design of the testbed with the network layout and how each device is connected to the physical process.

C. Testbed Operation

In our default process tank 3 is treated as the consumer with tank 2 as the supplier. When tank 3 is in the Filling state it will pump water from tank 2. When tank 2 is not full it will pump water from tank 1. Once tank 3 is full and tank 1 is empty it will empty completely into tank 1 before receiving water from tank 2 again. The testbed can be reprogrammed at any time.

III. Data Collection

The Historian regularly polls both PLCs and stores sensor and pin data with timestamps in an influxdb database. The central switch is configured with a mirror port which provides a copy of every packet passing through the switch for our network layer data collection. The data sets we have collected consist of three 6 hour "normal operation" process runs where our main process runs continuously with no faults or anomalies. We then recorded data sets from a number of different attack scenarios.

A. Attacks

The motivations of the attacks are to cause damage to the system or to the process. If a tank continues to fill beyond its set upper limit then the system will flood which could cause damage to the system or at best a loss of water. If a pump continues running when the input tank is empty this can damage the pump, additionally there can be a disruption in the logical representation of the process which can cause cascading failures or denial of service. In the following, we explain some of the different attacks executed on the SWaP testbed.

Sensor Pause
Record the value of the targeted sensor and continuously send that to the PLCs instead of updating with new readings.

Water Theft
While running the process as normal, remove water from the target tank. A pump with a flexible hose placed into the top of the tank was used to remove water.The water theft scenario adds a theft motivation as well as causing damage, it can also cause a denial of service by disrupting the balance of the process if there is no longer enough water in the system to fill a tank

Real Sensor Fault
This attack induces a fault at the physical. The target Arduino was turned off which prevented any sensor readings for that tank reaching the PLCs. This appears similar to the Sensor Pause attack but differs in the network traffic and the PLC sensor variables no longer updating instead of updating with the same value.

State Machine DoS
To "debounce" the water level readings in each tank there is a delay of several seconds during which the reading must remain below the "low level" value before the system will proceed to the next state. This is to prevent water sloshing from prematurely triggering the state transition. If the PLC level variable is above the limit for even one cycle it will restart the timer. As the historian only records PLC variables every second an attacker can regularly submit a fake value outside of that sample time which will stop the process from progressing through its state machine but the attack values will not be recorded by the historian's database (however the network traffic will show the attack packets.)

Stealthy Fake Sensor Inject
After an initial delay we take an average of recent water level changes and use this to craft fake readings that are a fraction of the real change. This results in the process appearing to proceed as normal while the water level passes the limits.

State Aware Stealthy Inject
The same as Stealthy Fake Sensor Inject but chooses which state of the state machine to start the attack in instead of a generic delay.

B. Publishing

The datasets we recorded will be made available for other researchers to download and run analysis on. We envision the combination of sensor readings and network traffic being particularly useful for those developing anomaly detection algorithms.

IV. Related Work and Motivation

At iTrust center in Singapore there are a couple of water system testbeds [1], [9], commissioning the fully functional water treatment and distribution testbeds from a third-party vendor with support from a public utilities board. In their system reconnaissance they discover a number of vulnerable configuration options made by their installer such as default and dictionary passwords. In [5] they perform a comprehensive series of attacks against all stages in their testbed and publish their dataset. However, those testbeds are costly to built and maintain and we were motivated to build a low cost and expandable testbed.

In [4] the team at Bristol Cyber Security Group discuss their growing CPS setup with an off-the-shelf water treatment plant, model factory and a building management system. They identify several requirements for an effective testbed including diversity of devices and software, scalability, complexity, data capture and student interaction.

There are some other efforts on ICS testbeds beyond water systems. An interesting effort in electric grid testbed is simulation-based Softgrid testbed [8] but there is no data generation and sharing. For an ICS testbed in CPS [7] authors highlighted that their prototype lacked the collection and distribution of data as it involves a manual process requiring time and resources. Another work in [3] presented simulated IEC61850 traffic and no information regarding the real process and the dynamics.

Previous research studies have tried to collect data from CPS settings but lack some desired features. We highlight a few of those,

- Simulated Data: Most of the datasets available are generated using simulated models [3]. Lack of real-world scenarios prompted us to do this work.
- Only Network Traffic Data: Previous data collection efforts only focused on the network traffic and as mentioned those were too simulated most of the time. One of the recent studies collected the network traffic from a realistic electric traction substation [6].
- Only Process Data: State of the art testbeds from iTrust [1], [9] record and share only process level data, without any information on the network layer data.

There have been efforts on building CPS testbeds for security research but very few were able to represent real-world scenarios and collect data to share with the academia and industry. We have focused on the process data from the sensors and actuators and tried to run the normal process for a range of different configurations.

A. Motivations

Access to the testbed results in impact in terms of access to resources and provides a lot of potential for running experiments and publishing datasets. There are several projects actively using the testbed as well as allowing students hands-on experience in ICS. The DIY nature allows greater understanding of every aspect of the system which is a huge help in planning experiments and for future expansion of the system. Building the testbed from low-cost consumer components not only frees up budget constraints but allows a wider range of choice and customisation especially when looking towards the future, we do not suffer from vendor lock-in or are constrained by earlier choices in the design process. Further it provides a flexibility in the range of attacks or experiments run as component damage is not a catastrophic loss and parts can be swapped out easily such as for fingerprinting.

V. Further Work

Currently the water flows in a simple loop, we would like to expand the number of stages and include processing stages such as filtration and a proper consumer stage. This would also allow us to add additional types of sensors and so more sources of data.

With a larger system we could also implement more sophisticated processes and add user control and feedback through a Human-Machine Interface (HMI) to more accurately represent real-world ICS and provide further attack scenarios. More stages would also allow additional models and manufacturers of PLCs to diversify our system and include more varied industrial protocols on the network.

Another aim is to set up a remote operational interface to allow other researchers access to the testbed to run their own experiments and collect their own data. Access to a physical system is often a challenge and we hope that making ours available will expand the amount of CPS research that can be done.

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